

Large Language Models in High-Volume Data Labeling and Predictive Classification: *A Study of Imminent Drug Relapse on Reddit*

Abstract

In the United States in 2020, over 41 million people aged 12 or older, needed substance use treatment ¹. Relapse is a common challenge in substance use recovery, with approximately 40–60% of individuals returning to substance use after a period of sobriety ². This study explores whether a machine learning model can predict impending drug relapse based on individuals' Reddit posting behavior in the month prior to a relapse event. We introduce a methodology that leverages large language models (LLMs) to rapidly annotate and organize a massive dataset of Reddit posts for this prediction task. Filtering from over 950,000 posts across multiple addiction-related subreddits, we identified 3,086 instances of self-reported relapse events using an iterative ChatGPT labeling pipeline. This high-volume labeling task, which would have required an estimated 20 years of manual effort, was completed in under 40 computer processor hours, illustrating the efficiency and feasibility of LLMs as automated data labelers. Engineered features reflecting common relapse triggers were also derived using LLMs and used to train various classifiers, including logistic regression, random forest, gradient boosting, SVM, and k-NN, to distinguish between collections of posts made four weeks versus one week before a users' relapse. Our results show that classical machine learning (ML) model performance remained near chance levels, suggesting that clear predictive signals were either absent or too nuanced for detection via these methods. In addition, we tested the LLM's ability to make direct predictions based on individual authors' posts, bypassing the classic ML models completely. This approach yielded more promising results and a level of potentially useful predictive accuracy. Our work highlights both the limitations of relapse prediction using Reddit content while also demonstrating the significant potential of LLMs for scalable behavioral data labeling, feature extraction, and as predictive classifiers.

Introduction

Drug addiction is a chronic relapsing condition, and predicting relapse is a longstanding goal for improving intervention and support. Relapse rates in substance use disorders are notably high, underscoring the need for tools to identify when someone is at risk of returning to use. In recent years, researchers have turned to social media data as a lens into individuals' mental states and behaviors. Online forums like Reddit host active recovery communities where people discuss their struggles, cravings, and victories in sobriety. This rich textual data offers an opportunity to observe potential warning signs of relapse in real time, if they can be reliably identified.

Prior studies have used Reddit and other social media to investigate substance use patterns, often relying on extensive hours of manual labeling in order to construct functional data sets on which to train predictive models. For example, Yang et al. [3] manually labeled relapse incidents on Reddit in order to test and train their GANS based predictive model. In the study titled 'Predicting Opioid Relapse Using Social Media Data' the authors described the labor required to label the data set as follows, "The dataset for this classifier, which consisted of 1,000 users (419 addicts and 581 non-addicts), took four people four days to compile as it required the researchers to read through all the comments made by each user and label them accordingly."

Similarly, Jha et al. [4] explored identifying changes in a user's posting content that might indicate a transition from recovery back to active use. These efforts suggest that language cues might foreshadow relapse, but they also rely on annotated data – i.e., knowing which users relapsed and when, so that predictive patterns can be learned.

Creating a labeled dataset of relapse events from social media is extremely labor-intensive. Relapse discussions are scattered among thousands of posts, and confirming a true relapse (with context like sobriety duration and timing) requires careful reading of posts, which can be many paragraphs long in some instances. Manually sifting through hundreds of thousands of posts for such rare events is a prohibitively time-consuming and costly bottleneck that has limited research in this and many other areas of research.

This study proposes a scalable alternative: using large language models (LLMs) as automated annotators. Recent advances in LLMs have shown that they can perform complex classification and labeling tasks with high accuracy⁵, effectively mimicking human understanding of text. We employed GPT4o-mini, GPT4.1-mini, and GPT4.1 in an iterative pipeline to scan and label Reddit posts, identifying those that describe a self-reported relapse by the post's author under specific criteria (detailed below). By using GPT to do the heavy lifting of reading and interpretation, we were able to construct a sizable dataset of confirmed relapse events in a fraction of the time a manual annotation would require. This approach demonstrates a new paradigm for behavioral research: harnessing AI to organize data that was previously too unwieldy to analyze.

In addition to the main labeling task (to isolate and label relapse events) we also used GPT4.1-mini to extract features from the final set of Reddit posts, such as common relapse triggers (stress, social pressure, cravings, etc.), to be used by ML models to help improve prediction accuracy.

With this curated dataset of relapse events, LLM extracted features, and the users' posting history, we then investigated our main research question: Can an individual's Reddit posting behavior predict if an upcoming drug relapse is imminent? Meaning, is the drug user about to relapse. Specifically, we compare each user's Reddit content four weeks prior to relapse with their content one week prior to relapse, to see if there are detectable differences that could serve as warning signals. To reiterate, our objective was not to predict whether a user will relapse in general, but to determine if a relapse is imminent within a short time window. To that end, we trained and evaluated multiple machine learning (ML) models using features derived from both LLM-generated labels and user text.

Finally, we explored the LLMs ability to directly predict the likelihood of relapse within 1 week or not using what we named the 'One Shot' labeling prompt. The One Shot prompt asked the LLM to express its confidence in the user's likelihood of imminent relapse on a scale of -5 to +5, with -5 being very unlikely, 0 being unclear, and +5 being very likely relapse is imminent.

Ultimately, our traditional ML model results were sobering: these models could not reliably distinguish posts made one week before a relapse from those made a month before. In other words, within the one-month timeframe, posting behavior did not contain clear predictive signals of an impending relapse that could be captured by generic features and traditional ML models. However, there was a limited level of success achieved by the LLM One Shot labeler. In short, by ignoring the unclear '0' labels and focusing on the LLM's most confident prediction (near -5 and near +5), we were able to isolate a subset of users on which prediction accuracy was over 70%.

We provide a detailed case study on using LLMs for data annotation in a complex, noisy domain – highlighting both the effectiveness (dramatically speeding up labeling) and the challenges (prompt engineering, error analysis) of this approach. We also make available a systematically gathered set of relapse-related posts which could facilitate qualitative analyses or future modeling attempts. Our work sits at the intersection of information science and substance use research, demonstrating how information-centric methods (natural language processing and machine learning) can be applied to problems in addiction and mental health. In the following sections, we describe our methodology, the LLM labeling pipeline, feature engineering, modeling experiments, results, and implications for future research.

Methods - Data Collection

We collected data from Reddit, a popular social media and discussion platform, in order to create two primary data sets. The first, a data set with identified drug relapse events, focused on communities in which people talk about addiction, drug use, and recovery. We targeted a set of subreddits including r/ Drugs, r/OpiatesRecovery, r/Addiction, and others related to various substances and recovery support (a complete list of subreddits is provided in the study's files). We chose a broad range of such subreddits to capture posts both from people actively using substances and those trying to remain sober. The second data set was the complete Reddit activity (across all subreddits) from the authors identified as having relapsed in the first data set.

Using complete archives of Reddit downloaded via Academic Torrents, we retrieved all posts made in the list of addiction related subreddits over a 13-month period (from December 2023 through December 2024). This timeframe was selected to cover one full year of data, which was recent enough to reflect current community dynamics and language. In total, our raw dataset for finding relapse events consisted of 957,396 posts, each with metadata including the author username, timestamp, subreddit, title, and the text content of the post. We did not collect comments, focusing only on top-level posts (original submissions by users) for this data set, under the assumption that a user's own posts would contain clearer, more detailed, and well-defined admissions of relapse.

Preprocessing: All posts were stored in a database and indexed by author and date. We performed light preprocessing on the text, such as lowercasing and stripping URLs or special characters, but avoided heavy filtering so as not to remove potentially relevant information (e.g., slang or misspellings related to drug use). We also noted that posts on Reddit can be of varying lengths; some are brief check-ins or questions, while others are long narratives of personal experience. We preserved the full text for the language model to analyze in context.

Identifying Relapse Events via LLM Annotation

Our first major task was to identify instances of drug relapse within the collected posts. We defined a "relapse event" based on the content of a post, using the following required criteria:

- **Self-reported relapse:** The post's author explicitly indicates that they themselves have used a substance after a period of sobriety (as opposed to talking about someone else's relapse).
- **Confirmed sobriety period:** The post (or context) makes clear that the author had been sober (abstinent) for at least 30 days before this use. This gives a clean window of 28 days which allows for comparing 4 weeks from relapse with 1 week from relapse (relapse imminent).
- **Recent, datable event:** The post describes a specific relapse incident that occurred recently (we aimed for within 2 weeks of the post-date OR an explicitly stated date of relapse by the author) and provides enough information to estimate the date of the relapse to within 3 days accuracy.
- **Expressed remorse or unintended use:** Typically, the tone of the post is that the author did not intend to slip up and is now disappointed or seeking help. We used this to filter out cases where someone might just be reporting routine use without regret (which would not be a "relapse" in the recovery sense).

Identifying posts that meet these criteria within a huge corpus is challenging. Instead of manual reading, we used various versions of OpenAI's GPT models to automatically label the posts. We designed an iterative prompt-engineering pipeline with five successive LLM annotation passes, each refining the selection:

1. **(Pass 1) Initial Relapse Filter** - GPT4o-mini: We prompted a GPT model to read each post and simply decide whether it describes a relapse by the author (True/False). This was a broad filter to catch any mention of using again after quitting. We used a relatively fast and inexpensive model variant (GPT4o-mini) for efficiency. This pass yielded 32,024 posts labeled as True (relapse event) out of 957,396. Posts labeled FALSE (not relapse event) were removed from the data set.
2. **(Pass 2) Sobriety Confirmation** - GPT4.1-mini: Many posts talk about using drugs but not all are relapses after long sobriety. In the second pass, we had GPT4.1-mini check each candidate from Pass 1 and verify that the author indicated at least 30 days of sobriety before the relapse. The prompt specifically asked the model to look for statements like "I was X days/weeks/months clean" or similar. This tightened the criteria and reduced the set to 10,584 posts that involved a relapse after 1 or more months of sobriety.
3. **(Pass 3) Context and Uniqueness Check** - GPT4.1-mini: Next, we further refined the labels by ensuring the posts truly fit a single, self-contained relapse event. The LLM was instructed to confirm: The post is about one clear relapse event (not multiple or vague references). The relapse is about the author themselves (not, say, "my friend relapsed" or a generic discussion about relapsing). The author's tone suggests it was an unintended slip with some remorse or guilt, distinguishing it from someone who has simply returned to regular use without regret. This pass eliminated false positives that slipped through earlier. After this filtering pass, 9,203 posts remained as high-confidence relapse event descriptions.
4. **(Pass 4) Date Extraction** - GPT4.1-mini: For each of these relapse posts, we wanted to assign an actual relapse date - a specific calendar date when the relapse happened. Often, posts describe the timing in relative terms (e.g., "I relapsed last night" or "on Saturday I used again"). We prompted GPT4 to read the post and output an estimated date of the relapse event (taking into account the post's timestamp for context). If a post said "relapsed yesterday" and it was posted on March 15, 2024, the model would output March 14, 2024 as the relapse date. We allowed a ± 3 day uncertainty in the estimate. This step reduced the dataset further, because in some posts the timing was too unclear to confidently assign a date. After date labeling, we had 4,279 relapse instances with associated dates.
5. **(Pass 5) Final Validation** - GPT4.1 (Full Model): The last pass used the full-strength GPT4.1 model (not the "mini" version) to double-check each of the remaining candidates. The model was given the post text and the previously

extracted information (sobriety length, estimated relapse date, etc.) and asked to confirm that: All the criteria were indeed met (self-reported, 30+ days sober, one recent event, etc.). The estimated relapse date was accurate given the post's clues.

Any posts that failed this rigorous check - for example, if the model realized the event was actually about someone else (not the post's author), or that the "30 days" of sobriety were not truly preceding the relapse date - were discarded. This final quality control yielded a definitive set of 3,384 relapse events. Each of these is a Reddit post wherein a user clearly confesses to a drug use relapse after 30+ days of sobriety, along with an approximate date of when that relapse occurred. A Python post-processor was used to cull redundant relapses (authors with 2 or more relapses in the entire 12-month span) down to the first relapse event only. Final relapse event count after the redundancy cull was 3,086.

Prompt Tuning Methods

Throughout this LLM labeling process, we took care to refine prompts on human reviewed sample sets (between 50 and 100 posts) and used a "rationale checked" version of each prompt prior to running the actual pass on the full data set. Meaning, during the prompt refinement process we included extra instructions dictating that the GPT model provide rationale for why it labeled a post a certain way. Reviewing these LLM explanations manually helped us catch systematic mistakes. In one example, when the model mistakenly flagged posts, the given rationale revealed that GPT was counting someone's cumulative sober time and incorrectly integrating this data to falsely estimate a relapse date. In another instance it was revealed that the LLM was attributing another person's (3rd party) relapse to the author. We clarified the prompt instructions based on such examples. Here is an example of updated prompt verbiage after initial tests of Pass 4 (GPT4.1-mini date labeler) in which the model had been confusing sobriety periods with estimating relapse dates... "Ignore unrelated durations—such as the author's total period of sobriety or addiction—unless they are clearly tied to when the relapse occurred." This iterative prompt tuning significantly improved precision in each subsequent pass. By the final pass, each GPT model was highly precise in identifying relapse narratives within our specified criteria.

Human Hours vs. LLM Hours

It is worth noting that using GPT in this manner dramatically accelerated what would otherwise have been an extremely time-consuming annotation task. In total, the five-pass pipeline took roughly 30–40 hours of straight computer/GPU processing time. This does not include human hours spent on prompt refinement and manual inspection of rationales for quality assurance, which took another approximately 40 hours of human oversight. By comparison, if a human annotator took even 2.5 minutes per post to evaluate relapse criteria, labeling ~950k posts would require about 40,000 hours (circa 20 person-years of work). Add to this the subsequent feature engineering labels (11 relapse indicators mentioned below) applied to the 72,000 weekly author comments and posts, and the total data preparation for the study would likely exceed 25 person-years of work. Not to mention the ability to pivot and alter the labeling criteria en masse, which would not be possible with human labeling. Thus, LLM-based annotation made this study possible where traditional methods would have failed or required a team of hundreds of human labelers.

Human Accuracy vs. LLM Accuracy

While it is obvious that using an LLM is faster than using a team of human labelers, it is very reasonable to ask "is it more accurate?" While this research does not delve directly into answering this question quantitatively, anecdotally it does appear that the LLM *can be* very much in the ballpark of human accuracy. The quality of the prompts created for the LLM is absolutely critical. A second factor, of potentially equal or even greater importance based on context, is consistency. The LLM has a level of consistency that cannot be matched by humans. A team of distinct unique humans would each individually view the labeling criteria in slightly different ways. They would all make different kinds of mistakes, judgments and estimations. In contrast, when using the LLM iteratively, the LLM is going to act extremely consistently and predictably across the entire data set. Although there can be variability in LLM responses when given the exact same input, the LLM does not get tired or change its perception over time. It seems very safe to assume an LLM labeler is far more consistent (not necessarily more accurate) than any single human labeler, let alone a team of labelers.

Note: we human/hand labeled from a subset just over 1000 entries that had initial LLM Passes applied. The process took over 40 hours and revealed the many nuances and grey area decisions that exist within the relapse criteria set for this study.

Producing the Data Set for Predictive Modeling

Having identified the 3,086 relapse events and their dates, the next step was to analyze the content of each user's posts, across the entirety of Reddit, over the 28 days prior to their respective relapse date. For example, if author1 relapsed on July 29th, 2024, we gathered all of author1's posts from July 1st up to and including July 28th. We excluded all posts on the relapse date itself, as this text would not be predictive/preceding the event, and would often contain text about the relapse itself.

This consolidated database was extracted from a complete data set containing all Reddit activity by all Reddit authors, across all Subreddits, not just authors found in this study. The expansive complete Reddit data set was downloaded via <https://academictorrents.com>. *Note: The number of authors was again reduced at this stage because not all authors who had admitted to relapses had prior Reddit activity. Many relapse admissions are made with 'throw away accounts', so that drug users can maintain anonymity. 2,193 authors remained after this data process.*

Once in the consolidated database the author data could be parsed and handled in various ways. Expressing all iterations of exploration would be exhaustive, so we will describe only a handful of explored permutations from our process.

Initial Exploratory Data Analysis (EDA)

To examine posting behavior relative to relapse timing, we segmented each user's activity into discrete weekly bins: 4 weeks, 3 weeks, 2 weeks, and 1 week prior to their self-reported relapse date. Specifically, these bins corresponded to days 28–22, 21–15, 14–8, and 7–1 before relapse. Additionally, we categorized user behavior by subreddit context: (1) all posts across all subreddits; (2) posts excluding addiction-related subreddits; and (3) posts limited to addiction-related subreddits. These segmentations were used to generate word clouds representing aggregate vocabulary as well as to produce weekly and daily sentiment trajectories using the VADER sentiment analysis algorithm. However, these visual analyses did not reveal any pronounced trends; overall sentiment remained largely stable across the 28-day pre-relapse period, regardless of subreddit category.

EDA based removal of outliers and culling resulting in final data set size.

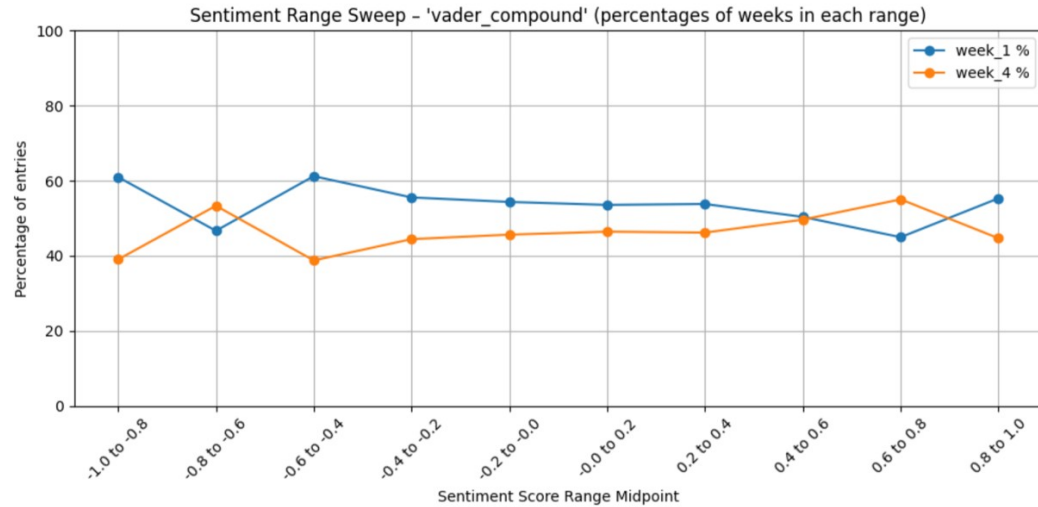
Additional entries were removed during the EDA process due to insufficient data or atypical posting behaviors. For example, authors who repeatedly submitted identical content — posting the exact same text multiple times — were excluded. Likewise, users who contributed an unusually large volume of posts were also removed. While these users may have been legitimate, the sheer quantity of their contributions risked disproportionately influencing the language models, potentially biasing the results. After these filters were applied, 1,664 authors remained for analysis.

Creation of Target Variable – “Relapse Imminent” (week_1 vs. week_4)

The central objective of this study was to determine whether a user actively attempting sobriety is approaching a relapse event. To operationalize this, we created a binary target variable labeled “relapse imminent.” For each author in the dataset, all Reddit posts made during the seven days immediately preceding their self-reported relapse (Week 1: days 1–7 prior) were concatenated into a single text entry and labeled as *relapse imminent* = TRUE (also labeled *week_1*). Similarly, all posts from the fourth week prior to relapse (Week 4: days 22–28 prior) were combined into a separate entry and labeled as *relapse imminent* = FALSE (also labeled *week_4*). This approach allowed us to frame the task as a binary classification problem: distinguishing the language of an individual one week from relapse versus four weeks out. Most users contributed posts in both *week_1* and *week_4*, yielding two labeled entries per author. However, in some cases, authors had relevant activity during only one of these time windows; no filtering was applied based on this criterion.

Initial Feature Analysis

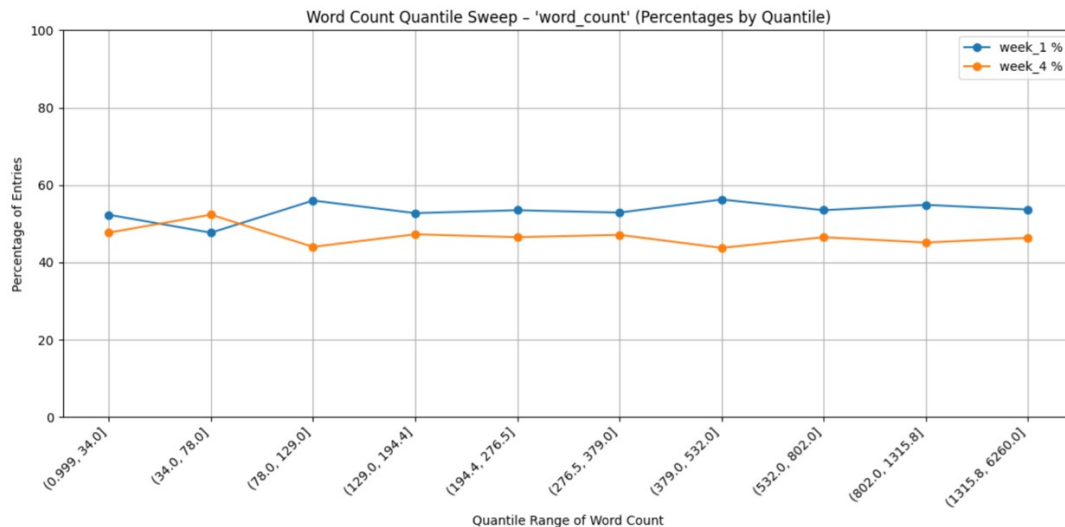
Word-Clouds: We viewed the text data from various perspectives to see if any patterns became readily apparent prior to training our machine learning models. The following two infographics represent word prevalence in *week_1* vs *week_4* posting content. There are very minor differences but overall, there do not seem to be any clear differences in the aggregate text 1 week prior to relapse vs. 4 weeks prior to relapse.



[86]:

	range	midpoint	pct_week_1	pct_week_4	Total in Range
0	-1.0 to -0.8	-0.9	60.975610	39.024390	41
1	-0.8 to -0.6	-0.7	46.666667	53.333333	30
2	-0.6 to -0.4	-0.5	61.224490	38.775510	49
3	-0.4 to -0.2	-0.3	55.555556	44.444444	144
4	-0.2 to -0.0	-0.1	54.353562	45.646438	379
5	-0.0 to 0.2	0.1	53.566959	46.433041	799
6	0.2 to 0.4	0.3	53.810264	46.189736	643
7	0.4 to 0.6	0.5	50.364964	49.635036	274
8	0.6 to 0.8	0.7	44.961240	55.038760	129
9	0.8 to 1.0	0.9	55.319149	44.680851	94

Word Count Analysis: As with sentiment analysis, word count does not present any obvious correlation with the week_1 or week_4 entries. The chart below, similar to the sentiment analysis chart, creates word count bins based on 10 quantile ranges. The first bin has entries with 1 to 34 words for a given author's week worth of posts (week_1 or week_4), the second bin has entries 34 to 78 words for a given author's week worth of posts, and so on ending with authors that posted between 1,315 and 6,260 words in a given week. As with word-clouds and sentiment analysis, we did not see any obvious predictive value in the word count feature. *Note: creating percentile ranges results in splits at the decimal level, however there is only ever a whole number value of words per week per author entry.*



Preliminary ML Model Results - Week 1 vs. Week 4 (3 Features)

With the dataset now organized, we proceeded to test whether we could predict if an author was near relapse from their posting behavior using traditional machine learning (ML) models. We formulated this as a classification task. Each relapse event usually yields two data points or entries. The first is an author's cumulative posts at 1 week prior to relapse, and the second is an author's cumulative posts at 4 weeks prior to relapse. As mentioned earlier, not all authors had posts present in both week 1 and week 4. For the ML models, we labeled the 1-week-prior instance as class 1 (imminent relapse coming within a week) and the 4-weeks-prior instances as class 0 (not an imminent relapse). The model's job is to distinguish between these two classes based on the features of the posts. Essentially, we ask: given a complete set of posts from a random week in a recovering user's life, can we tell if this week is right before their relapse or still a month out? If the model can't do better than chance, it implies the content in those periods is indistinguishable in terms of relapse risk signals.

NOTE: The ML modeling data set, containing and comparing only week_4 and week_1 entries, contained 1,546 authors across 2,582 entries. Of the 1,664 authors in the main data set, 118 posted only in week_3 and or week_2 and were no longer present in our predictions. In addition, not all authors posted in both week_4 and week_1, hence a total 2,582 entries instead of the 3,092 (1,546 x 2) that one might expect to see.

Machine Learning Models Utilizing 3 Features (Author Text, VADER Sentiment, Word Count):

- Logistic Regression: A simple linear model to see if a weighted combination of triggers could linearly separate the classes.
- Random Forest Classifier: An ensemble of decision trees, which can capture nonlinear interactions between triggers.
- Gradient Boosting Machine (e.g., XGBoost or LightGBM): An advanced tree-ensemble method often good at structured data.
- Support Vector Machine (SVM): To see if a maximal margin separator in feature space performs better, possibly with an RBF kernel for nonlinearity.
- K-Nearest Neighbors (k-NN): A non-parametric approach to check if instances naturally cluster by class in feature space.
- Neural Network (Keras): A deep learning model used to capture complex, nonlinear interactions among features that may signal relapse risk, particularly when traditional models underperform.

Result: No Predictive Models Found

Unfortunately, no models were able to produce significant predictions from the original set of features. Models hovered between 48% and 52% accuracy. Exact accuracy numbers were not relevant as accuracy variations were driven more by changing data split random seeds than by changing hyper parameters and tuning the models themselves. These results remained consistent across a number of permutations. For instance, we experimented with different week combinations Week 3 vs. Week1, Week 4, vs Week2, etc. which had no noticeable impact. The addition of VADER sentiment analysis, word counts, and various text vectorization methods were also inconsequential. The apparent truth was that a balanced formation of the data was reading as noise to all our trained models.

Feature Engineering with LLMs

After the failure to gain predictive value from the initial feature set (Author Text, VADER Sentiment, Word Count), we worked to build new features derived from the author text data. There were 2 distinct LLM feature engineering operations used to create new potentially predictive features from the text data. Our goal was to extract features that might indicate rising risk or warning signs and even admissions that a relapse was imminent.

Operation 1 – Drug Relapse ‘Trigger Labels’ (Per-Post – 72,645 Entry Iterations)

The trigger labeling was done as a per-post operation. Meaning, of the 72,645 posts made by the authors, across week 4 and week 1, each post was given 11 new labels (circa 800k total labels applied). The first 9 being 'trigger labels', defined as text related to known drug relapse triggers, such as interpersonal conflict, emotional stress, social pressure, etc. In addition, there was a label for direct admissions or declarations by the author that they felt near relapse or at high risk of relapse. Lastly, included was a label in which the LLM observed 'protective factors' present, such as signs of positive coping, support, involvement in recovery, etc.

The TRUE/FALSE trigger labels were then combined into counts and percentages at the week level. For instance, if Author1 had 5 posts labeled with interpersonal conflict across their 50 total posts in Week 4, the newly engineered Week 4 labels would be `interpersonal_conflict_count: 5` and `interpersonal_conflict_percentage: 0.1`.

The exact 11 directives given to the LLM iterator for each label in our batch processor is as follows:

direct_relapse_reference - The author in some way states that they are presently concerned that they themselves are about to relapse, or are about to enter a situation which could very likely trigger a relapse.

emotional_stress - expressions of anxiety, depression, frustration, irritability, hopelessness, shame, guilt, loneliness, or emotional overwhelm.

interpersonal_conflict - signs of relationship problems such as arguments, family tension, breakups, betrayal, custody battles, or being misunderstood.

social_pressure_peers - mentions of being encouraged or tempted to use by friends, coworkers, party settings, or peer groups known for substance use.

environment_cues - being near places or objects associated with past substance use (e.g., bars, neighborhoods, drug paraphernalia, certain smells or routines).

loss_or_grief - references to the death of loved ones, divorce, estrangement, or emotional fallout from major losses.

boredom_or_lack_of_purpose - mentions of aimlessness, lack of structure, boredom, or feeling like there's "nothing to do" or "no reason to stay clean."

financial_stress - indicators of job loss, debt, eviction risk, money anxiety, or lack of means to meet basic needs.

physical_pain_or_fatigue - complaints of chronic pain, illness, injury, physical burnout, poor sleep, or extreme tiredness.

substance_craving - direct or implied desire to use a specific drug, romanticizing past use, or thoughts like "just once" or "I miss how it made me feel."

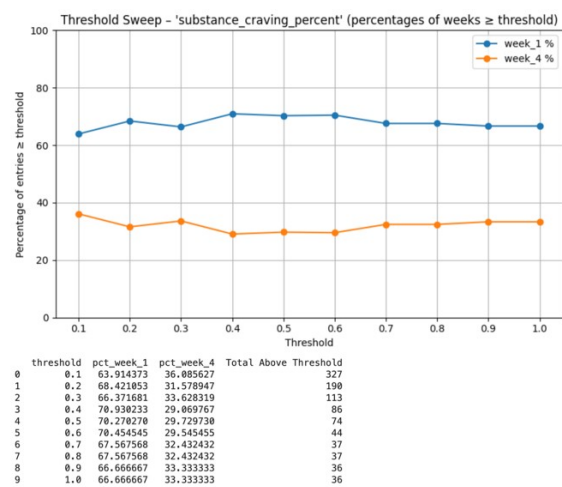
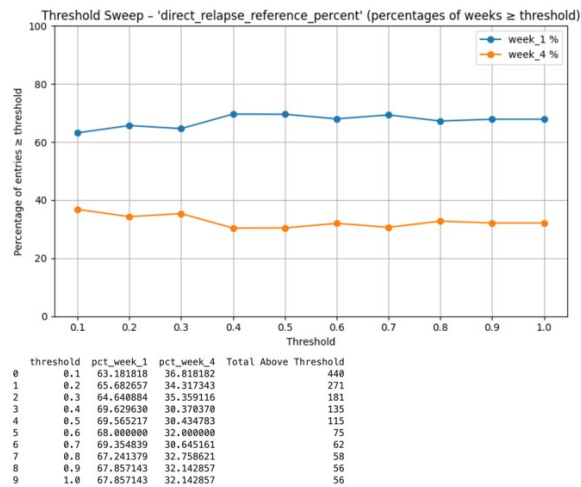
protective_factors_present - presence of sobriety support, recovery goals, therapy, accountability partners, stable living conditions, self-care habits, or spiritual connection.

Reviewing Potential Feature Value/Predictiveness

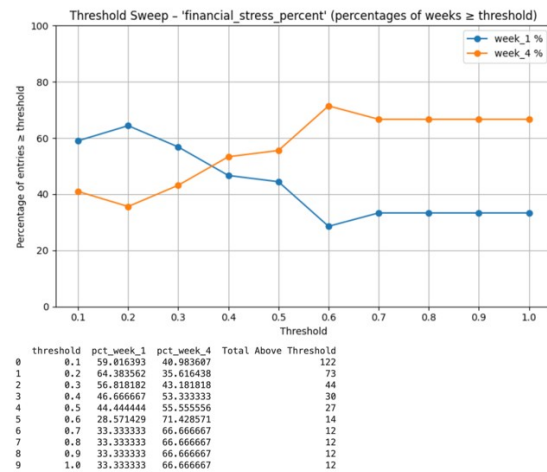
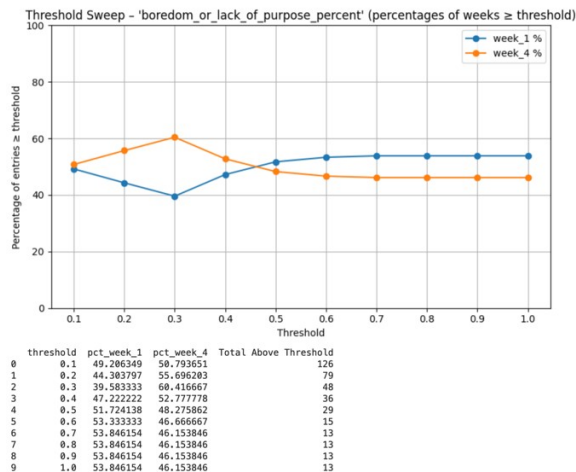
To gain insight as to the predictive value of each feature, we graphed the percentage of entries associated with week_1 vs week_4 when looking at an increasing percentage of feature presence. The charts below show how often entries with a threshold from 0.1 through to 1 are associated with week_1 or week_4 entries.

In the first chart, `direct_relapse_reference_percent`, we can see that when looking at all entries that have a 0.1 or greater value that there are 440 entries that fit this criteria. Of these 440 entries, 63.18 percent are week_1 entries and 36.82 percent are week_4 entries. Meaning, the metric is predictive. However, as we increase the threshold for this feature, predictive value does not progressively and consistently increase in tandem as we might hope.

NOTE: Many authors only have 1 or 2 posts in an entire week. For this reason, we often see a value of 1 (100% of posts are TRUE for a given feature). In other words, low post counts can skew the data.



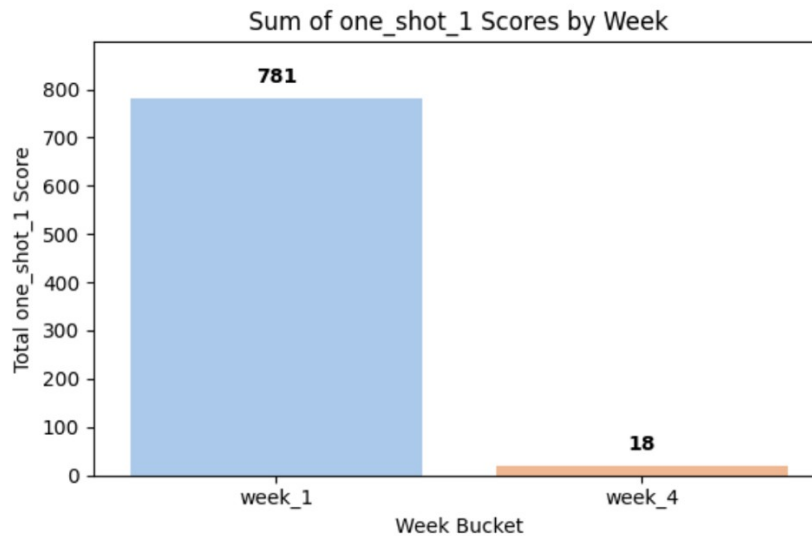
Above we see that 'direct relapse reference' and 'substance craving' have notable predictive value. This is made clear not only by the separation between the week_1 and week_4 lines on the graph, but also in the high number of entries that meet the criteria (Total Above Threshold). Below we can see that 'boredom or lack of purpose' and 'financial stress' are not reliable predictive features. These 2 features have erratic and inverting relationships, as well as lower 'Total Above Threshold' numbers, indicating that a low number of entries meet the threshold criteria.



Operation 2 - 'One-Shot' Relapse Likelihood Prediction (Per-Week - 2,582 Entry Iterations)

The 'One-Shot' label was created in addition to the previous trigger labels. This label is a direct attempt to see if an LLM could do the work of our traditional ML predictive modeling when given the entire weeks-worth of posts from each author as an input. We prompted GPT4.1-mini to provide a single rating on a scale from -5 to +5 answering: "How likely does it seem, from reading this content, that the author will relapse in the very near future?" A rating of -5 would mean "very confidently NOT going to relapse soon" (perhaps the user seems very securely in recovery), whereas +5 would mean "extremely likely to relapse imminently" (perhaps the user is expressing intent to use or severe struggle). Ratings at or near 0 would indicate neutral or unclear.

We intended this as a kind of summary score: essentially asking GPT4.1-mini to be a human-like judge of the posts, to see if even an intelligent reader might sense an upcoming relapse or not. We did this for the 4-weeks-prior and 1-week-prior texts separately. A quick review of the sum totals of the One-Shot labels suggests that the LLM was able to make predictions, at least in aggregate. The week_1 (near relapse) sum was 718, while the week_4 (far from relapse) posts had a sum of only 18. As the week_1 predictions were positive numbers and the week_4 predictions were negative numbers, we would have hoped to see a negative value in the week_4 bucket, but the predictive value is still clear in that we are seeing a much higher week_1 sum.



The bar graph also suggests, as is also shown in metrics later in this paper, that the LLM was more capable of predicting week_1 entries than predicting week_4 entries. This shortcoming/imbalance may be due to the quality of the prompt given to the LLM and or the nature of making predictions in the drug relapse space more generally.

Machine Learning Model Results - Week 1 vs. Week 4 (13 Features)

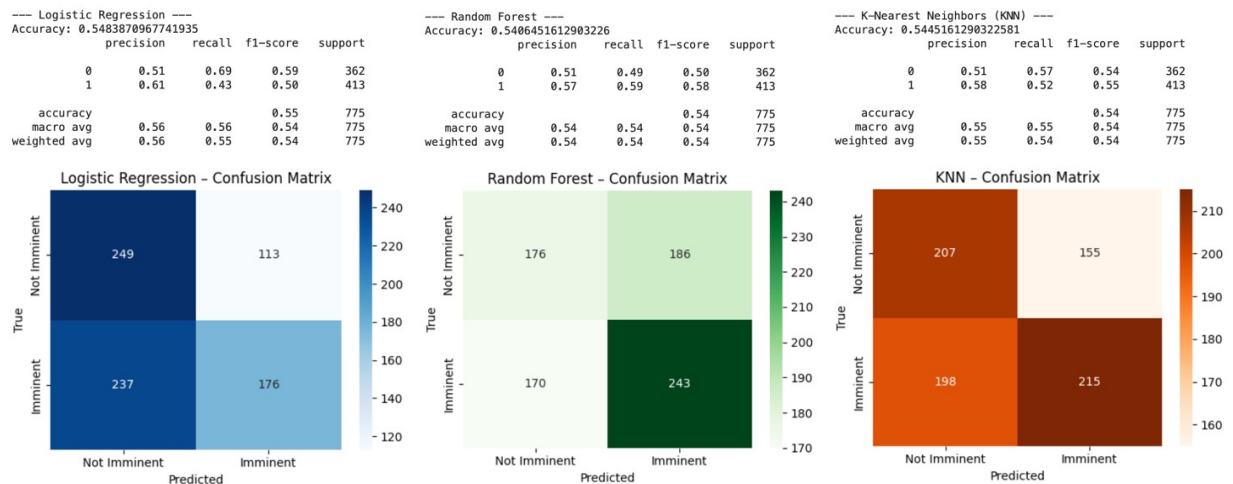
After outlier filtering, the data set for the final ML model builds contained 1,546 authors with 2,582 entries. This author-to-entry ratio means that 510 authors posted only in Week 4 or Week 1, but not in both weeks. 1,036 authors had posting behavior present in both Week 4 and Week 1. While prediction accuracy was not strong, we were able to get slightly more consistent results than we did from our initial 3-feature-models. The new feature set supported models that often landed around 54% predictive accuracy. The best models used the following 13 features:

- vader_compound
- one_shot_1
- direct_relapse_reference_percent
- emotional_stress_percent
- interpersonal_conflict_percent
- social_pressure_peers_percent
- environment_cues_percent
- loss_or_grief_percent
- boredom_or_lack_of_purpose_percent
- financial_stress_percent
- physical_pain_or_fatigue_percent
- substance_craving_percent
- protective_factors_present_percent

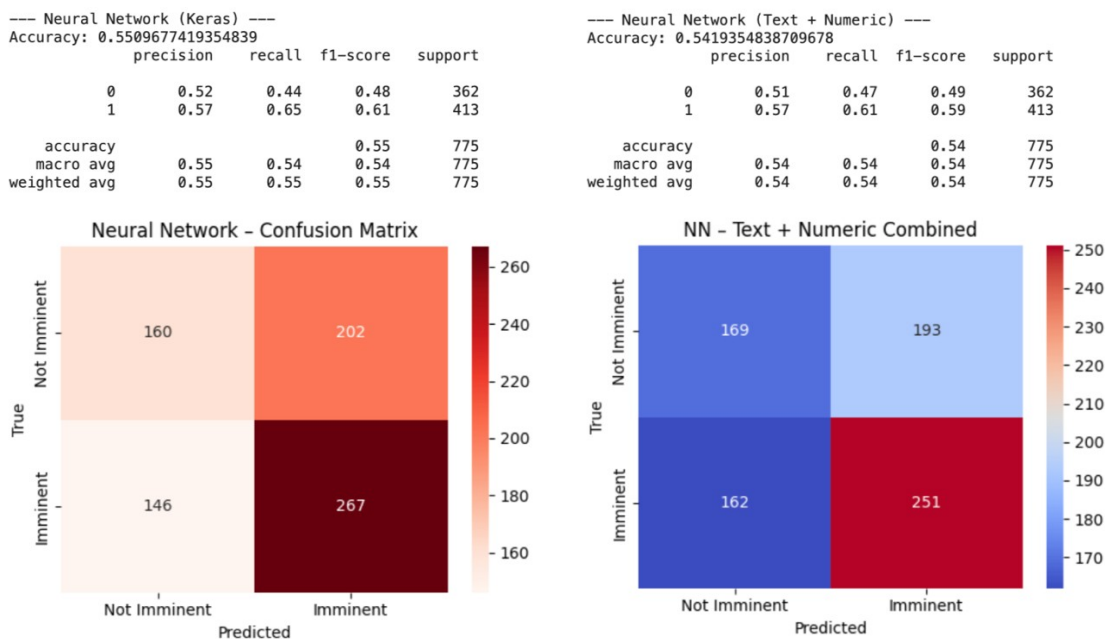
We split the data into 70% training and 30% testing sets, training and evaluating using the same set of ML models as were used in the preliminary model tests. It is worth noting that unlike the initial 3-feature-models, these 13-feature-models were far less sensitive to altering the random seed of the training test splits and often had less than a 1% shift when switching random seeds.

We trained each model on the training set and tuned hyperparameters via cross-validation where applicable (for example, the number of trees for RF/GBM). Performance was measured primarily by classification accuracy and F1-score for distinguishing class 1 vs class 0. We also looked at the accuracy and recall scores with an accompanying confusion matrix for each model.

The printed results and confusion matrices below show the outcomes for our 3 strongest traditional ML models.



Our neural network models (NN) were not any more successful than the traditional ML models. The results for each iteration were hovered between 52% and 55% on each training.



The above matrix on the left was run on the 13-feature-set. The matrix on the right was trained on both the 13-feature-set as well as the original text, reintegrated in vectorized form.

Result: No Highly Predictive Models Found (Although, slightly better accuracy compared to initial models)

Even with the addition of the new LLM generated features, we found no substantial predictive value in the ML models we trained. Furthermore, the data set was imbalanced with week_1 containing 1377 entries (53.3%) and week_4 containing 1,205 entries (46.7%). We also trained models utilizing balanced tests splits (test split contained exact 50/50 split of week_1 and week_4 entries) with results generally falling around 52%/53% accuracy. In this scenario, there could be various arguments for balancing the test split or leaving it as-is, but in our case neither option creates a worthwhile outcome.

Direct LLM Predictions Explored - Week 1 vs. Week 4 (One Shot Labels)

For the last step in our research, we extended our exploration and evaluation of the LLM One Shot Labels. As noted in our initial EDA, the One Shot labels showed aggregate predictive value by producing an expected higher sum in the week_1 vs. week_4. We decided to see what the LLM's prediction accuracy would be when removing all entries in which the LLM gave a '0' prediction. Based on the One Shot prompt, '0' is the label the LLM applied when it felt it could not predict either way if the author was near or not near relapse. The total number of '0' entries was 1280, just shy of half of the total entries in the data set. The remaining 1302 entries had prediction labels ≥ 1 & ≤ -1 . The accuracy on this subset of entries, on which the LLM made some kind of prediction, was 57.5%. The complete outcome metrics are shown below with an accompanying confusion matrix.

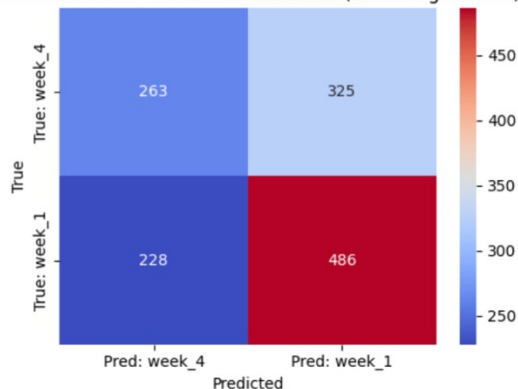
```
Number of rows used: 1302
Accuracy: 0.5752688172043011

Classification Report:
              precision    recall  f1-score   support

     0         0.54         0.45         0.49         588
     1         0.60         0.68         0.64         714

 accuracy          0.57         0.56         0.58         1302
 macro avg          0.57         0.56         0.56         1302
 weighted avg          0.57         0.58         0.57         1302
```

Confusion Matrix - Threshold Prediction (Excluding Neutral)



Given the results shown by omitting the '0' (no confidence) entries, we then decided to run a series of progressively tighter filtering constraints, moving ever towards the LLM's most confident prediction ends of +5 and -5. The general accuracy results were as follows.

Predicting using positive and negative ends of the LLM's prediction spectrum.

- ≥ 1 & ≤ -1 filter creates a subset of 1302 entries with accuracy at 57.5% (Excludes only '0' labeled predictions)
- ≥ 2 & ≤ -2 filter creates a subset of 1088 entries with accuracy at 58.5% (Excludes -1 to +1 labeled predictions)
- ≥ 3 & ≤ -3 filter creates a subset of 873 entries with accuracy at 60%

- ≥ 4 & ≤ -4 filter creates a subset of 588 entries with accuracy at 63%
- ≥ 5 & ≤ -5 filter creates a subset of 196 entries with accuracy at 62.8%

Isolating positive predictive value only (All predictions are week_1 exclusively)

- ≥ 1 filter creates a subset of 811 entries with accuracy at 60%
- ≥ 2 filter creates a subset of 653 entries with accuracy at 62%
- ≥ 3 filter creates a subset of 519 entries with accuracy at 66%
- ≥ 4 filter creates a subset of 345 entries with accuracy at 71%
- ≥ 5 filter creates a subset of 95 entries with accuracy at 72.6%

Isolating negative predictive value only (All predictions are week_4 exclusively)

- ≤ -1 filter creates a subset of 491 entries with accuracy at 53.5%
- ≤ -2 filter creates a subset of 435 entries with accuracy at 52.6%
- ≤ -3 filter creates a subset of 354 entries with accuracy at 51.9%
- ≤ -4 filter creates a subset of 243 entries with accuracy at 51.4%
- ≤ -5 filter creates a subset of 101 entries with accuracy at 53.4%

A useful subset relying more heavily on positive predictions.

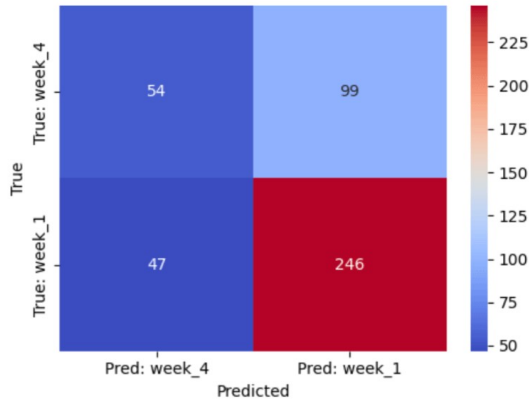
- A ≥ 4 & ≤ -5 filter creates a subset of 446 entries with accuracy at 67%

As we can see from the isolated positive and negative filters, the LLM was far more accurate in predicting week_1 entries. This means when using the LLM to purely predict only week_1 (relapse imminent), the LLM had higher accuracies, with accuracy starting around 60% and progressively climbing as high as 72.6%. Conversely, setting the positive value to null and using only negative prediction, we see that the LLM was less accurate when predicting only week_4 (relapse NOT imminent), with accuracies remaining between 51.4% and 53.5%. Below are the complete metrics for two filtered subsets that feel particularly useful in their compromise between high accuracy and high entries remaining (Number of rows used).

Number of rows used: 446
Filter: ≥ 4 & ≤ -5
Accuracy: 0.672645739910314

Classification Report:				
	precision	recall	f1-score	support
0	0.53	0.35	0.43	153
1	0.71	0.84	0.77	293
accuracy			0.67	446
macro avg	0.62	0.60	0.60	446
weighted avg	0.65	0.67	0.65	446

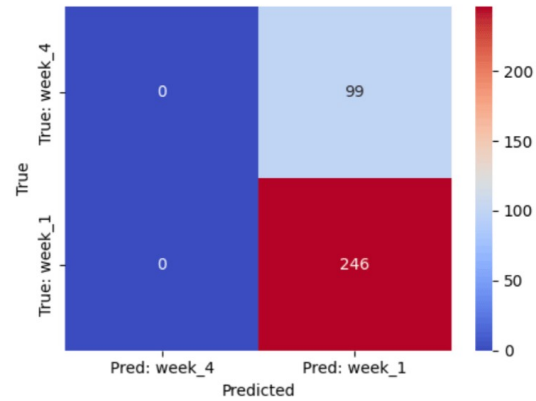
Confusion Matrix – Threshold Prediction



Number of rows used: 345
Filter: ≥ 4
Accuracy: 0.7130434782608696

Classification Report:				
	precision	recall	f1-score	support
0	0.00	0.00	0.00	99
1	0.71	1.00	0.83	246
accuracy			0.71	345
macro avg	0.36	0.50	0.42	345
weighted avg	0.51	0.71	0.59	345

Confusion Matrix – Threshold Prediction



The outcome displayed on the above left utilizes entries with predictions of 5, 4, or -5 only. It contains 446 entries (17% of the complete data set) and boasts an accuracy of 67%. The results on the above right utilizes entries with a prediction of 4 or 5 only. This yields a result with 345 entries (13% of the complete data set) and an accuracy of 71%. As is made clear in the confusion matrix, all predictions were of week_1 in this subset.

Result: Strong Predictive Potential

We believe, based on our evaluation of the LLM's One Shot results, that there is a great deal of predicative power that can be derived directly from LLM's when applied to conversational/language based data. Unlike conventional machine learning workflows, which typically follow structured linear pipelines, applying an LLM for direct prediction feels unfamiliar yet can be both viable and robust for language-based data problems. By removing low-confidence predictions (those labeled '0' by the LLM), the resulting subset showed a modest accuracy of 57.5%, which improved steadily as filtering was tightened toward the model's most confident predictions. Notably, when isolating highly positive scores (e.g., ≥ 4), accuracy climbed to over 70%, while the model's ability to predict negative (non-relapse) cases remained comparatively weak and hovered near chance levels. These findings suggest that LLMs can offer meaningful predictive power directly from text, especially in identifying relapse-imminent states, and that confidence-weighted filtering may offer a valuable method for leveraging LLM outputs in clinical or behavioral risk assessment contexts.

Ethical Considerations

While this study leverages publicly available Reddit data to investigate relapse prediction, we acknowledge the sensitive nature of the content, which often involves deeply personal disclosures. All analyses were conducted using anonymized data, and no usernames or direct quotes are included in this work. Our intent is to study behavioral patterns at scale, not to stigmatize or identify individuals. Importantly, although the use of LLMs for data analysis and prediction opens new possibilities, we caution against applying such models in real-time monitoring or intervention systems. Flagging individuals as "relapse imminent" based on social media behavior raises significant ethical concerns, including privacy violations and the risk of unintended harm. These tools should be used strictly for retrospective research and hypothesis generation unless future deployment is guided by robust ethical frameworks and informed consent protocols.

Discussion

Prediction in the Drug Relapse Space Proved Difficult (Especially with Traditional ML Models)

One of the primary findings of this research is that predicting short-term relapse from Reddit posting behavior appears to be extremely challenging, at least with the kind of features and approach we used. This negative result is important to report, as it tempers expectations about what social media data alone can tell us in the context of relapse. There are several factors to consider in interpreting this outcome:

1. **Lack of Explicit Signals:** Many might assume that people give off some clear warning signs before a relapse – for example, talking about cravings, or expressing hopelessness, or disengaging from the recovery community. While such signs do occur for some individuals, our data shows they are not reliably or universally present. A substantial portion of users did not discuss their urges or struggles on Reddit in the lead-up to using again. Some may have kept those feelings private or only shared them in offline settings (like with a sponsor or friend), not on a public forum. Others might not have experienced a long buildup at all; relapse can happen suddenly (e.g., running into an old friend, one bad decision, etc., without a long period of planning or public anguish). Thus, the absence of strong signals in text could simply reflect the reality of how relapses often occur.
2. **Noisy and Diverse Content:** Reddit posts range widely in topic. On any given week, a person in recovery might post directly about their sobriety, recovery or events that likely relate to a relapse, however they may also post about any number of unrelated events and topics that exist across the complete set of all subreddits used in this research. Furthermore, a user could be internally struggling. The models only see what is written. If the critical risk factors are more internal (mood, craving) and not voiced, the model has little to latch onto.
3. **Timing and Window Size:** We chose 4 weeks vs 1 week as the comparison based on the hypothesis that a month-out and just-about-to-relapse might differ. It could be that this timeframe is not optimal. Some relapses might show signals earlier (e.g., a pattern that started changing 2+ months out), or conversely only in the final days or hours (shorter than a week). Our week-long aggregation might also blur short-lived crises. For instance, perhaps 3 days before relapse something was different, but averaging over 7 days or mixing with other posts could hide it. A more fine-grained temporal analysis (like daily posting patterns, or looking at inter-post intervals) might reveal subtle cues that our current features missed.
4. **Features and Modeling Limitations:** We relied on versions of GPT4 to extract a set of conceptually relevant features (triggers and sentiments). It's possible that these features, while intuitive, do not capture the full linguistic nuances that precede relapse. A more direct text modeling approach (such as training a neural network on the raw text sequences of posts) could theoretically pick up on complex patterns that our handcrafted features lost. However, given the small data size of the final set of relapse authors fitting our criteria and the high variability, training such a model would be difficult.
5. **Individual Differences:** Relapse is a very personal phenomenon, influenced by each individual's circumstances, substance of choice, coping skills, support network, etc. What one person's relapse trajectory looks like can be different from another's. Our model was trying to find common denominators across people, which might be an ill-posed objective. It could be more fruitful to focus on personalized modeling – i.e., for a given individual, detect changes from their own baseline behavior that might signal relapse risk (a form of anomaly detection in their posts). We did not do that here because it requires enough historical data per person and was outside our scope, but it is a potential direction. In practice, interventions are often tailored to each person's triggers; a one-size-fits-all model from social media might simply not exist.

LLM's for Research Data Labeling

Despite limited predictive success, our study has a significant methodological contribution: the use of LLMs to perform large-scale data labeling and feature extraction iteratively. This proved to be a game-changer in feasibility. The fact that GPT4 could read hundreds of thousands of posts and accurately identify relapse events with rich contextual criteria is remarkable. Traditional content analysis or even crowdsourced annotation would have been painfully slow and likely inconsistent on such a task. We demonstrated that with careful prompt design and iterative refinement, LLMs can emulate expert human judgment (in our case, essentially acting as a domain expert who understands the nuances of recovery and relapse) to an extent that is useful for building research datasets. Previous work has noted LLMs achieving high agreement with human annotators on various

tasks ⁵, and our experience aligns with that – GPT4 was able to apply our specific complex definition of “relapse post” quite reliably after prompt tuning.

We also showed LLMs can handle open-ended feature coding. Instead of manually coding for triggers, GPT4 effectively did a content analysis for us, across thousands of text segments, in a consistent manner. This approach could be extended to other mental health contexts – for instance, detecting markers of depression or self-harm in social media posts – where we can encode expert-based categories and let the AI annotate a large corpus. There are certainly caveats: the model’s understanding might not be perfect and could introduce bias or errors. One must still perform quality checks and perhaps small-scale validations with human experts. But as a force multiplier, LLM annotation can accelerate the research pipeline dramatically.

LLM’s for Direct Prediction

Perhaps the most notable outcome of this study was the unexpected utility of the LLM’s direct prediction scores generated during the One Shot labeling task. Although this avenue of investigation was not part of the original study design, we developed the single-shot prompt as a final exploratory step after conventional ML models failed to demonstrate reliable performance. The prompt was informed by prior experiments in label generation, though it lacked an established metric to guide iterative improvement. Despite these limitations, the LLM’s prediction outputs displayed meaningful predictive accuracy, suggesting that LLMs may possess an intrinsic capacity to detect relapse risk signals in text. These results indicate strong potential for future research focused on refining prompt engineering and implementing structured prompting pipelines to enhance predictive performance.

Future Research

Drug Relapse Future Research:

Given our findings, we have several recommendations for researchers aiming to predict drug relapse or behavior outcomes from social media or textual data:

- Consider broader timeframes or alternate targets: For relapse prediction, instead of trying to predict a relapse within a very short window, it might be easier to identify people at risk of relapse at all in the future, versus those who will maintain recovery. That would involve having a comparison group of users who did not relapse (which our study lacked since we only focused on those who did). A classification between “will eventually relapse vs will not” might glean signals that “imminent relapse vs not” did not.
- Use richer data modalities: Reddit posts are just one source of data. Incorporating other signals – such as frequency of posting (did they suddenly stop posting?), linguistic changes over time (using more first-person singular, or different vocabulary), or even data from other platforms (if available) – could improve prediction. Some studies have looked at sentiment trajectories or social engagement changes as predictive features.
- Personalized alert systems: As mentioned, tracking individuals against their own baseline might detect when something is off. For example, if a user usually posts very encouraging content and suddenly one week their posts are pessimistic and mention struggles, that deviation might be more telling than absolute content. LLMs could be used in a personalized way to flag “unusual tone” for a specific user.
- Community-level signals: Perhaps individual prediction is too noisy, but aggregate patterns could exist (for example, if the entire community’s discourse shifts, or if a person’s interactions with others change). Network analysis or community sentiment analysis might reveal something at a higher level of abstraction.

LLM usage for Future Research:

Based on our model outcomes and our experiences throughout our research processes, we see two areas with the greatest potential for LLM implementation in future research.

1. Using LLMs as Data Labelers

Our research demonstrated the powerful utility of LLMs in automating the data labeling process—an innovation that has the potential to revolutionize how labeled datasets are constructed in resource-constrained or domain-specific research environments. Future research could further explore the standardization and reproducibility of LLM-based labeling across varying domains (e.g., healthcare, finance, education, legal, etc.), ensuring consistency in interpretation even across nuanced or subjective text. A promising direction involves integrating LLMs into active learning frameworks, where the model is used to pre-label data and human reviewers validate only the most uncertain cases, thus minimizing human effort while maximizing label quality.

In addition, there is strong potential in using LLMs for dynamic and context-aware feature generation. Instead of relying solely on fixed keyword dictionaries or static ontologies, LLMs can be used to infer latent behavioral or emotional features based on text content. These engineered features can be evaluated not just qualitatively, but quantitatively in terms of their predictive power when incorporated into downstream models. Future work should also explore benchmarking frameworks to systematically compare LLM-generated labels to traditional human annotations and to evaluate their effect on downstream model performance in classification or regression tasks.

2. Using LLMs Directly as Predictive Models

The potential of LLMs to serve as predictive engines—rather than merely as components in a data preparation pipeline—presents a significant shift in the traditional ML paradigm. In our limited exploration, we found that LLMs exhibited notable predictive capabilities directly from natural language inputs when coupled with prompt engineering. Future research could investigate the systematic development of prompt optimization algorithms, including reinforcement learning from model feedback (RLMF), retrieval-augmented prompting, or prompt ensembling methods to enhance prediction consistency and accuracy. These methods could help refine prompts over iterative cycles, potentially improving model output beyond what traditional fine-tuning offers.

Additionally, LLMs may be embedded as predictive nodes within larger machine learning systems, functioning in parallel with traditional models or as pre-screeners. Research should explore how to modularize LLM prompts within end-to-end pipelines, allowing for scalable deployment where LLMs act as task-specific evaluators of unstructured inputs (e.g., user posts, clinical notes, or open-ended survey responses). Evaluation frameworks could include comparative studies on performance, interpretability, and cost-efficiency versus conventional supervised models.

Finally, there's also room to explore hybrid architectures, where structured model features are supplemented by LLM-inferred judgments. These could take the form of soft predictions (e.g., probabilities) or structured outputs (e.g., ranked categories or causal statements). Research into post-hoc calibration of LLM outputs and integration with model ensembling techniques could unlock further robustness, especially in domains requiring high reliability and explainability.

Limitations

We acknowledge several limitations in our study. First, our data relied on self-reports of relapse on Reddit, which likely underestimates actual relapse occurrences and potential includes trolling or fake relapse events. Some people may relapse and not post about it at all, or they might leave the community out of shame, etc. Our findings thus apply to those who do share about their relapse. Second, the accuracy of our LLM labeling, while high, is not perfect. It's possible some events in our dataset are misclassified or the dates slightly off. This could add noise that hampers modeling. Third, the sample size of 1,546 relapse events (authors), while a good number for a rare event, is still relatively small for training complex models, especially given the diversity across events. Fourth, we treated all substances and all individuals together; in reality, relapse dynamics might differ (relapse to alcohol vs opioids might have different typical precursors, for instance). We did not stratify by substance or other demographics due to data sparsity, but that could be a factor. Finally, our feature set, though extensive, might have missed other potential indicators (e.g., changes in writing style, or social support responses from others).

Conclusion

In summary, this study attempted to predict imminent drug relapse in individuals based on their Reddit postings in the month leading up to a relapse. We introduced a novel methodology using iterative LLM pipeline to rapidly label relapse events and extract meaningful features from a large volume of online data, demonstrating a significant advancement in research tooling for social media analysis. The assembled dataset allowed us to conduct a focused investigation into behavioral signals preceding relapse.

Our key finding is that Reddit content alone did not yield a successful predictive classical machine learning model for imminent relapse while in contrast, use of an LLM for making direct predictions appears promising. The differences between posts made four weeks before and one week before relapse was too subtle and inconsistent for classical machine learning models and, within the scope of our limited research, difficult even for an advanced LLM to reliably discern. These results highlight the complexity of relapse as a phenomenon and suggests caution in assuming that social media users will outwardly exhibit detectable signs of an impending relapse.

The process of reaching this conclusion provided valuable insights. We showed that with careful design, LLMs can function as effective large-scale data labelers, overcoming what used to be a prohibitive data curation hurdle. This opens the door for researchers to study other nuanced behaviors or rare events in online communities that were previously impractical to capture at scale. The study also underscores the importance of sharing null results; knowing that a particular approach did not work can save others time and guide the field toward more promising directions.

For practitioners and community moderators, an implication is that automatically flagging people who might relapse soon (based on their posts) is not straightforward with current methods – direct human support and intervention likely remain irreplaceable. However, the methods developed here could assist in monitoring trends or identifying relapse reports quickly, which could be useful for outreach.

Future work should explore more quantitative evaluations of LLM prompts geared towards direct prediction to further the goal of predicting and preventing relapse. Moreover, the ethical dimension of such predictive efforts should be kept strongly in focus – any system that attempts to foresee a mental health crisis or relapse must be handled with care to avoid harm.

In conclusion, the marriage of information science techniques and addiction research – exemplified by the use of LLMs and machine learning on Reddit data – is a promising avenue to generate new insights. We hope this work inspires others to harness modern AI for social good, while also reporting candidly on the challenges and limitations encountered.

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